

LAC 1.2

$$\langle u, v \rangle = u^T A v$$

$\langle \cdot \rangle$ je skalarowy iloczyn $\Leftrightarrow$ A jest symetryczny; $a_{11} > 0$; $\det A > 0$.

⊖ A jest symetryczny: $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

$$a_{11} > 0 \text{ bo } a > 0 \text{ a } \det A > 0 \quad a c - b^2 > 0$$

$$\Rightarrow \langle u, v \rangle = \langle v, u \rangle$$

$$\langle u, v \rangle = a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2$$

$$\langle v, u \rangle = a v_1 u_1 + b v_1 u_2 + b v_2 u_1 + c v_2 u_2 \quad (\text{z komutacyjności } + \text{ nie zmieniaj})$$

ii)

$$\langle u, t v \rangle = u^T A (t v) = t u^T A v = t \langle u, v \rangle$$

$$iii) \langle u, v + w \rangle = u^T A (v + w) = u^T A v + u^T A w = \langle u, v \rangle + \langle u, w \rangle$$

$$iv) \langle u, u \rangle = a u_1^2 + 2b u_1 u_2 + c u_2^2$$

$$\text{z } a c - b^2 > 0 \Rightarrow c > \frac{b^2}{a} > 0 \Rightarrow c > 0; \sqrt{a} \sqrt{c} > b$$

z powyższych: 1.) $u_1 u_2 \geq 0$; potem

$$\langle u, u \rangle = \underbrace{(\sqrt{a} u_1 - \sqrt{c} u_2)^2}_{\geq 0 \text{ (kwadrat)}} + \underbrace{2(\sqrt{a} \sqrt{c} u_1 u_2 + b u_1 u_2)}_{\geq 0 \text{ z } \sqrt{a} \sqrt{c} > b}$$

2.) $u_1 u_2 \leq 0$, potem

$$\langle u, u \rangle = (\sqrt{a} u_1 + \sqrt{c} u_2)^2 - \underbrace{2\sqrt{a} \sqrt{c} u_1 u_2}_{\geq 0} + 2b u_1 u_2$$

a bo $2\sqrt{a} \sqrt{c} u_1 u_2 + 2b u_1 u_2 \geq 0$

$$\langle u, u \rangle = 0 \Leftrightarrow u = \vec{0} \quad (\text{C jest kwadrat})$$

jeżeli $u \neq \vec{0}$; wtedy $u_2 = t u_1$, potem

$$\langle u, u \rangle = a u_1^2 + 2b t u_1^2 + c t^2 u_1^2 = 0 \quad (t^2 + 2b t + a = 0)$$

$$\text{bo } (2b)^2 - 4ca \geq 0; 4b^2 - 4ac \geq 0; ac - b^2 \leq 0$$

(z dyskryminantą)

jeżeli $\Rightarrow \det A > 0$

($\Rightarrow$) $\langle u, v \rangle = u^T A v$ je sledovni minor

1.) $\langle u, v \rangle = \langle v, u \rangle$.

($\hookrightarrow$) $u^T A v = \underline{v^T A u}$

$\hookrightarrow = (v^T A u)^T = u^T A^T v$

(leba je to čisto)

$\Rightarrow u^T A v = u^T A^T v$; $u^T A v - u^T A^T v = 0$; $u^T (A - A^T) v = 0$;

$A - A^T = 0$; $A = A^T \Rightarrow A$ je simetrična $\rightarrow A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

2.) $\hookrightarrow$ pozitivni definitnost nam je $\langle u, u \rangle > 0$ za $u \neq 0$

od $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ tedaj $\langle u, u \rangle = a(1)^2 + 2b(1)(0) + c(0)^2 = a$

$\langle u, u \rangle > 0$; tedaj $a > 0$.

3.) polno $u = \begin{pmatrix} 1 \\ t \end{pmatrix}$; $\langle u, u \rangle = a + 2bt + ct^2 \underset{m}{\geq} 0$

to nastane kot diskriminanta $\Delta < 0$;

$(2b)^2 - 4ac < 0$; $4b^2 - 4ac < 0$; $a - b^2 > 0$

$\Rightarrow \det A > 0$.